Prayas JEE AIR (2025)

WEEKLY TEST 22

DURATION : 180 Minutes | **DATE : 21/12/2024** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1.	(3)
2.	(1)
3.	(2)
4.	(3)
5.	(3)
6.	(4)
7.	(2)
8.	(3)
9.	(3)
10.	(3)
11.	(1)
12.	(1)
13.	(2)
14.	(2)
15.	(1)
16.	(1)
17.	(2)
18.	(2)
19.	(4)
20.	(2)
21.	(3)
22.	(1)
23.	(20)
24.	(11)
25.	(3)

CHEMISTRY

26.	(4)
27.	(1)
28.	(4)
29.	(1)
30.	(2)
31.	(3)
32.	(3)
33.	(1)
34.	(2)
35.	(3)
36.	(1)
37.	(1)
38.	(1)
39.	(2)
40.	(3)
41.	(2)
42.	(2)
43.	(2)
44.	(3)
45.	(2)
46.	(85)
47.	(4)
48.	(9)
49.	(9)
50.	(165)

MATHEMATICS

51.	(1)
52.	(1)
53.	(4)
54.	(2)
55.	(4)
56.	(1)
57.	(4)
58.	(1)
59.	(3)
60.	(2)
61.	(1)
62.	(4)
63.	(1)
64.	(1)
65.	(4)
66.	(3)
67.	(3)
68.	(4)
69.	(1)
70.	(3)
71.	(2)
72.	(3)
73.	(64)
74.	(6)
75.	(3)

SECTION-I (PHYSICS)

1. **(3)**

$$\sqrt{\frac{2\ell}{a_2}}-\sqrt{\frac{2\ell}{a_1}}=t \Rightarrow \frac{\sqrt{2\ell}}{t}=\frac{\sqrt{a_1a_2}}{\sqrt{a_1}-\sqrt{a_2}}$$

$$\sqrt{2a_1\ell}-\sqrt{2a_2\ell}=v \Rightarrow \frac{\sqrt{2\ell}}{v}=\frac{1}{\sqrt{a_1}-\sqrt{a_2}}$$

$$\Rightarrow \frac{v}{t}=\sqrt{a_1a_2} \Rightarrow v=\sqrt{(a_1a_2)t}$$

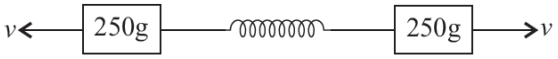
2. **(1)**

$$U=-\frac{Kq^2}{d}+QV$$

$$V=-\frac{KP}{D^2}=-\frac{Kqd}{D^2};\ U=-\frac{Kq^2}{d}-\frac{KQqd}{D^2}$$

3. **(2)**

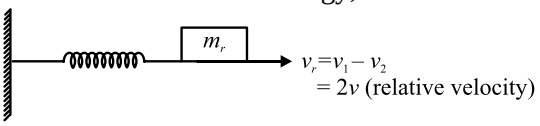
Let m = 250g = 0.25kg



Reduced mass of the system

$$m_r=\frac{m_1m_2}{m_1+m_2}=\frac{mm}{m+m}=\frac{m}{2}$$

From conservation of energy,



$$K.E_i+P.E_i=K.E_f+P.E_f$$

$$\frac{1}{2}m_rv_r^2+0=0+\frac{1}{2}kx^2$$

$$=\frac{1}{2}\times\frac{m}{2}\times4v^2=\frac{1}{2}kx^2$$

$$\frac{kx^2}{2}=mv^2$$

$$\frac{2}{2}x^2=0.25v^2$$

x = 0.5v

$$x=\frac{v}{2}=\frac{1}{4}\text{m}=25\text{cm}$$

4. **(3)**

$$B_P=\left(\frac{\mu_0 i}{4r}+\frac{\mu_0 i}{4\pi r}\right)=\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$$

5. **(3)**

We have,

$$\vec{F}=5\hat{i}+3\hat{j}-7\hat{k},\ \vec{r}=2\hat{i}+2\hat{j}+\hat{k}$$

$$\vec{\tau}=\left|\vec{r}\times\vec{F}\right|\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 2 & 1\\ 5 & 3 & -7\end{vmatrix}=17\hat{i}+19\hat{j}-4\hat{k}$$

6. **(4)**

At focus, magnification is ∞.

7. **(2)**

Elongation, $\Delta L=\frac{FL}{AY}\Rightarrow\Delta L\propto\frac{1}{Ay}$

$$\therefore\ \frac{\Delta L_A}{\Delta L_B}=\left(\frac{A_B}{A_A}\right)\left(\frac{Y_B}{Y_A}\right)=\left(\frac{3}{1}\right)\left(\frac{4}{1}\right)=\frac{12}{1}$$

8. **(3)**

Reflected wave will be 3.1 cos [1.8z + (5.4 × 10^6)t]$\hat{i}$ *N/C*, comparing it with

$$E_0=\cos\left(Kz+\omega t\right)\hat{i}\,\frac{N}{C}$$

K = 1.8

$$\frac{2\pi}{\lambda}=1.8$$

$$\lambda=\frac{2\pi}{1.8}=\frac{\pi}{0.9}=\frac{10\pi}{9}=3.48\,\text{m}$$

9. **(3)**

For Monatomic

$$\frac{C_p}{C_v}=\frac{5}{3}$$

Triatomic

$$\frac{C_p}{C_v}=\frac{4}{3}$$

For Diatomic

Rigid $\frac{C_p}{C_v}=\frac{7}{5}$

Non Rigid $\frac{C_p}{C_v}=\frac{9}{7}$

10. **(3)**

$$^{218}_{84}A\xrightarrow{\alpha}{}^{214}_{82}A_1\xrightarrow{\beta^-}{}^{214}_{83}A_2\xrightarrow{\gamma}{}^{214}_{83}A_3\xrightarrow{\alpha}$$

$$^{214}_{83}A_3\xrightarrow{\alpha}{}^{210}_{81}A_4\xrightarrow{\beta^+}{}^{210}_{80}A_5\xrightarrow{\gamma}{}^{210}_{80}A_6$$

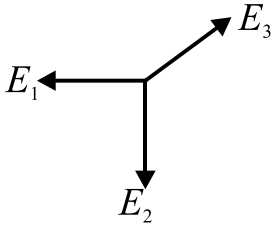
210 and 80 is the correct answer.

11. **(1)**

$$E_1=\frac{kq}{l^2}=E_2$$

$$E_3=\frac{kq}{\left(\sqrt{2}l\right)^2}=\frac{kq}{2l^2}$$

$$E=\frac{kq}{2l^2}\left(2\sqrt{2}-1\right)$$



12. **(1)**

$N = mv^2/r$

$\Rightarrow$ The graph given in option (a) suits the best for the above relation.

13. **(2)**

At steady state, capacitors are open.

$$\therefore i = \frac{4V}{6+2+8} = \frac{1}{4}A$$

Voltage across C_1 is

$$V_1 = V_{AC} = i(6\Omega + 2\Omega) = 8i \text{ Volt}$$

Voltage across C_2 is

$$V_2 = V_{BD} = i(2\Omega + 8\Omega) = 10i \text{ Volt}$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{8i}{10i} = \frac{4}{5}$$

14. **(2)**



$$x_{cm} = \frac{1}{M}\int_0^L x.dM$$

$$dM = \lambda.dx = \left(a + b\left(\frac{x}{L}\right)^2\right).dx$$

$$x_{cm} = \frac{\int x dM}{\int dM} = \frac{\int x\lambda dx}{\int \lambda dx} = \frac{\int_0^L x\left(a + \frac{bx^2}{L^2}\right)dx}{\int_0^L \left(a + \frac{bx^2}{L^2}\right)dx}$$

$$= \frac{a\left(\frac{x^2}{2}\right)_0^L + \frac{b}{L^2}\left(\frac{x^4}{4}\right)_0^L}{a(x)_0^L + \frac{b}{L^2}\left(\frac{x^3}{3}\right)_0^L}$$

$$= \frac{a\frac{L^2}{2} + b\frac{L^2}{4}}{aL + \frac{bL}{3}} = \frac{(2a+b)L}{(3a+b)4} \times 3$$

$$= \frac{3L}{4}\left(\frac{2a+b}{3a+b}\right)$$

15. **(1)**

According to the faraday's law of electromagnetic induction, $EMF = \frac{-d\phi}{dt}$

When magnet passes through the centre rogion of loop, no emf is induced. By using the Lenz's law, the polarity of emf will be negative while entering the loop.

16. **(1)**

Universal law of gravitation holds good for any pair of bodies in the universe. This is correct statement.

At the centre of the earth gravitational acceleration is zero. Therefore, the weight of any person becomes zero when the person is at the centre of the Earth. i.e., Weight = mg = 0

17. **(2)**

$$\sqrt{(2d)^2 + (d)^2} - 2d = \lambda$$

$$\Rightarrow \left(\sqrt{5} - 2\right)d = \lambda \Rightarrow d = \frac{\lambda}{\left(\sqrt{5} - 2\right)}$$

18. **(2)**

$$\rho = \rho_0\left(1 - \frac{r^2}{R^2}\right), 0 < r \le R$$

As, sphere is floating, so weight of liquid is balanced by buoyancy force.

$\Rightarrow$ mg = B

$$\Rightarrow \int \rho(4\pi r^2 dr)g = \rho_L \frac{4}{3}\pi R^3 g$$

$$\Rightarrow \int_0^R \rho_0 4\pi\left(r^2 - \frac{r^4}{R^2}\right)dr = \rho_0 4\pi\left(\frac{r^3}{3} - \frac{r^5}{5R^2}\right)_0^R$$

$$\Rightarrow \rho_L = \frac{2}{5}\rho_0$$

19. **(4)**

$$\because E_n = -\frac{13.6Z^2}{n^2}eV$$

$$\Rightarrow \Delta E = 13.6Z^2\left[\frac{1}{2^2} - \frac{1}{4^2}\right]eV$$

$$= 13.6 \times (4)2\left(\frac{1}{4} - \frac{1}{16}\right)eV$$

$$\Rightarrow \Delta E = 40.8eV$$

20. **(2)**

We know that for SHM, potential energy is minimum of mean position and is maximum at its extreme position. Also, the sign of potential energy is the same throughout the SHM.

21. **(3)**

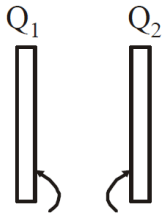
$$3 \times \frac{v}{4l_c} = 4 \times \frac{v}{2l_0} \text{ or } \frac{l_c}{l_0} = \frac{3v}{4} \times \frac{2}{4v} = \frac{3}{8}$$

22. (1)

$$V = \frac{Q}{C}$$

$$= \left(\frac{Q_1 - Q_2}{2C}\right)$$

$$= \left(\frac{4-2}{2\times 1}\right) = 1\text{V}$$



$$\frac{Q_1 - Q_2}{2} - \frac{(Q_1 - Q_2)}{2}$$

23. (20)

As $V(t) = 220 \sin \frac{50}{3}\pi t$

so, $I(t) = \frac{220}{50}\sin\frac{50}{3}\pi t$

i.e., $I = I_m = \sin\frac{50}{3}\pi t$

For $I = I_m$

$$t_1 = \frac{\pi}{2\omega} = \frac{3}{100}\text{sec}$$

and for $I = \frac{I_m}{2}$

$$\Rightarrow \quad \frac{I_m}{2} = I_m \sin\left(\frac{50}{3}\pi t_2\right)$$

$$\Rightarrow \quad \frac{\pi}{6} = \frac{50}{3}\pi t_2$$

$$\Rightarrow t_2 = \frac{1}{100}\text{s}$$

$$\therefore \quad t_{\text{req}} = \frac{3}{100} - \frac{1}{100} = \frac{2}{100} = \frac{20}{1000}\text{s} = 20\,\text{ms}$$

24. (11)

Using, intensity $\text{I} = \frac{\text{nE}}{\text{At}}$

n = no. of photoelectrons

$$\Rightarrow 16\times 10^{-3} = \left(\frac{\text{n}}{\text{t}}\right)\times\frac{10\times 1.6\times 10^{-19}}{10^{-4}} \text{ or, } \frac{\text{n}}{\text{t}} = 10^{12}$$

So, effective number of photoelectrons ejected per unit time = $10^{12} \times 10/100 = 10^{11}$

25. (3)

For slab in series, we have

$$\text{R}_{eq} = \text{R}_1 + \text{R}_2 = \frac{x}{\text{KA}} + \frac{4x}{2\text{KA}} = \frac{3x}{\text{KA}}$$

Now, in a steady state rate of heat transfer through the slab is given by

$$\frac{\text{dQ}}{\text{dt}} = \frac{\text{T}_2 - \text{T}_1}{\text{R}_{\text{eq}}} = \frac{(\text{T}_2 - \text{T}_1)}{3\text{x}}\text{KA} \qquad \text{…..…(i)}$$

$$\text{Given } \frac{\text{dQ}}{\text{dt}} = \left(\frac{\text{A}(\text{T}_2 - \text{T}_1)\text{K}}{\text{x}}\right)\text{f} \qquad \text{…...(ii)}$$

Comparing (i) and (ii), we get f = 1/3

SECTION-II (CHEMISTRY)

26. (4)

Energy required for 1 Cl_2 molecule $= \frac{242\times 10^3}{\text{N}_\text{A}}$

Joules. This energy is contained in photon of wavelength λ

$$\frac{\text{hc}}{\lambda} = \text{E}$$

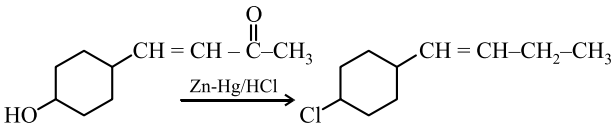
$$\Rightarrow \quad \frac{6.626\times 10^{-34}\times 3\times 10^8}{\lambda} = \frac{242\times 10^3}{6.022\times 10^{23}}$$

$$\lambda = 4947\,\text{Å} = 494\,\text{nm}$$

27. (1)

ZnHg/HCl can't be used due to the presence of acid sensitive group i.e. OH

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and Na/Liq. NH_3 and $NaBH_4$ convert – CO – into – CH(OH)–

28. (4)

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$

$$\text{K}_\text{c} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = 64$$

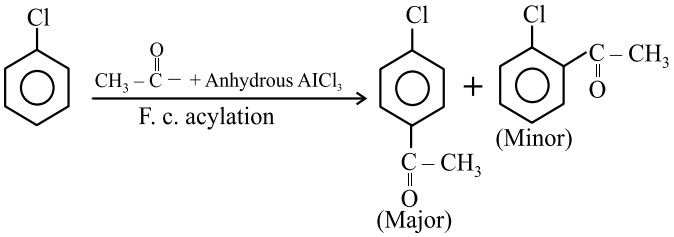
The reaction is reversed and multiplied by $\frac{1}{2}$, then for reaction,

$$\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g})$$

$$K_c' = \frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]} = \frac{1}{\sqrt{K_c}} = \frac{1}{\sqrt{64}} = \frac{1}{8}$$

[5]

29. **(1)**



Chlorine is ortho/para directing, para is major.

30. **(2)**

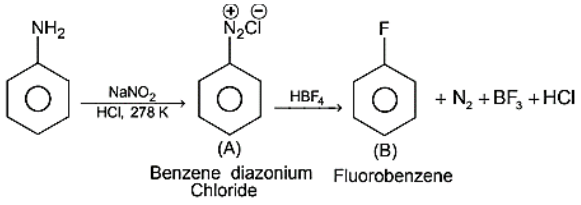
$\overset{+6}{C}rO_4^{-2} \rightarrow Cr^{+3}$ Change in oxidation number =3

$\overset{+7}{M}nO_4^- \rightarrow Mn^{+2}$ Change in oxidation number =5

$\overset{+6}{C}_2O_7^{2-} \rightarrow 2Cr^{+3}$ Change in oxidation number =6

$\overset{+3}{C}_2O_4^{-2} \rightarrow 2\overset{+4}{C}O_2$ Change in oxidation number =2

31. **(3)**



32. **(3)**

$\because \quad Fe^{3+} + 3e^- \rightarrow Fe; E^\circ = -0.036\,V$

$\therefore \quad \Delta G_1^\circ = -nFE = -3F(-0.036) = +0.108\,F$

Also $Fe^{2+} + 2e^- \rightarrow Fe; E^\circ = -0.439\,V$

$\therefore \quad \Delta G_2^\circ = -nFE = -2F(-0.439) = 0.878\,F$

To find E° for $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$

$\Delta G^\circ = -nFE = -1FE^\circ$

$\because \quad G^\circ = G_1^\circ - G_2^\circ$

$\therefore \quad G^\circ = 0.108\,F = 0.878\,F$

$\therefore \quad -FE^\circ = +0.108\,F - 0.878\,F$

$\therefore \quad E^\circ = 0.878 - 0.108 = 0.77\,V$

33. **(1)**

Total Node = n – 1

Radial Node = n – l –1

Angular Node: l

Here, l is Azimuthal Quantum number

Here, Angular node =0

$\Rightarrow$ l = 0 i.e s orbital

Radial nodes =2

$\Rightarrow$ n – l – 1 = 2

n – 0 – 1 = 2 $\Rightarrow$ n = 3

Hence, orbital is 3 s

34. **(2)**

Molecules	Oxidation state of **N**
NO	+2
NO_2	+4
N_2O	+1
NO_3^-	+5

Hence, order of oxidation state is $NO_3^- > NO_2 > NO > N_2O$

35. **(3)**

(1) Lucas reagent → Conc. HCl and $ZnCl_2$

(2) Dumas method →CuO/Δ

(3) Kjeldahl's method → Conc. H_2SO_4/Δ

(4) Heinsberg reagent →$C_6H_5SO_2Cl$/ aq. KOH

36. **(1)**

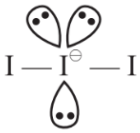
Along the period, metallic radius decreases and atomic mass density increases.

So, the correct order is:

Cu > Co > Fe > Cr > Zn

37. **(1)**

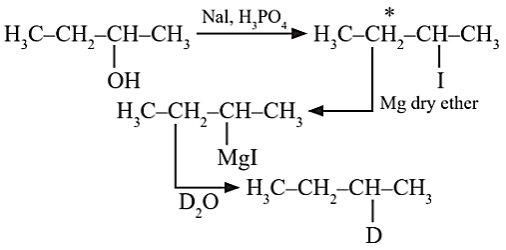
The structure of I_3^-



Linear (180°), sp^3d hybridisation is:

38. **(1)**

In the first step of the reaction substitution will takes place and in the second step of the reaction formation of grignard reagent will takes place after with hydrolysis by D_2O will takes place.



39. **(2)**

Atomic radius of boron is smaller than that of beryllium. Hence IV statement is false. Rest are true statements.

40. **(3)**

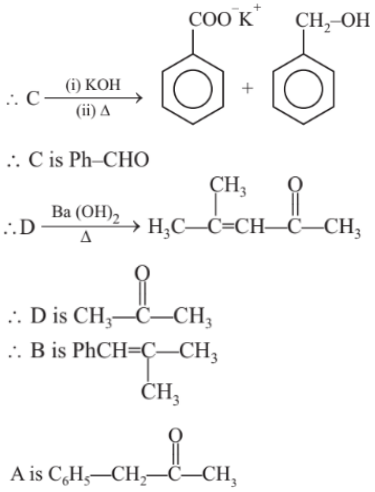
Maltose is a disaccharide which is formed by forming C_1–C_4 glycosidic linkage between two D- glucose, molecules

41. (2)

At infinite dilution, molar conductivity is measured which is also called limiting molar conductivity (λ_m^∞). Thus, both strong electrolyte (KCl) and weak electrolyte (CH_3COOH) have same value of conductivity at infinite dilution stage. Hence, statement-I is not true.

Statement-II is also not true because Λ_m increases by decreasing the concentration of electrolyte.

42. (2)



43. (2)

(i) Sn–HCl

(ii) Fe–HCl

(iii) Zn–HCl

Can convert nitrobenzene to aniline

44. (3)

$\Delta G^\circ = -2.303RT \log K_{cq}$

$= -2.303 \times 8.314 \times 298 \log 10^{-14}$

$= -2.303 \times 8.314 \times 298 \times -14$

$= 79,881.87$

$\simeq 80 KJmol^{-1}$

45. (2)

Given $K_3[Co(CN)_6]$ is inner orbital complex with hybridization d^2sp^3 and octahedral geometry. Ligands are approaching metal along the axes. Hence, $d_{x^2-y^2}, d_{z^2}$ orbitals are directly in front of the ligands.

46. (85)

Given, $\alpha = 0.4$,

For dissociation

$i = 1 + (n-1)\alpha$

$i = 1 + 4\alpha = 2.6$

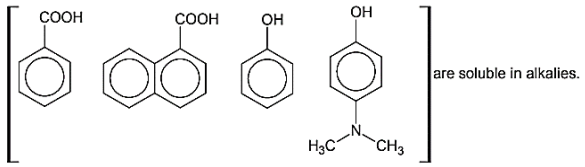
$(\Delta T_b) K_4[Fe(CN)_6] = (\Delta T_b)$ solution

$i \times k_b \times m = i \times k_b \times m$

$2.6 \times k_b \times 1 = 1 \times k_b \times \frac{18.1 \times 1000}{M \times 81.9}$

$M = 85.$

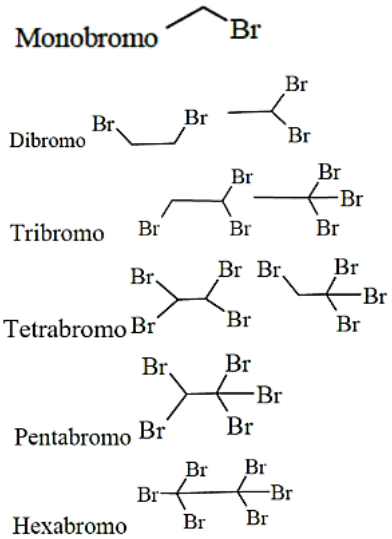
47. (4)



48. (9)

Bromination of ethane gives the following products.

$CH_3 - CH_3 + + Br_2 \xrightarrow{h\nu}$



49. (9)

$H_3C - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} = CH - CH_3$

50. (165)

Rate constant for first order reaction is given as,

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{70 \times 60} = \frac{6930}{7 \times 6} \times 10^{-6} = 165 \times 10^{-6}\ s^{-1}$$

SECTION-III (MATHEMATICS)

51. (1)

The equation $x^2 + x + 1 = 0$ has complex roots of unity 1, ω, ω^2.

$$A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$$

$$A^2 = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Now, $A^{31} = A^{28} \cdot A^3 = I \cdot A^3 = A^3$

52. (1)

S1: Given $A^T = A, B^T = -B, C^T = -C$

Let $M = A^{13}B^{26} - B^{26}A^{13}$

Then $M^T = \left(A^{13}B^{26} - B^{26}A^{13}\right)^T$

$= \left(A^{13}B^{26}\right)^T - \left(B^{26}A^{13}\right)^T$

$= \left(B^T\right)^{26}\left(A^T\right)^{13} - \left(A^T\right)^{13}\left(B^T\right)^{26}$

$= B^{26}A^{13} - A^{13}B^{26} = -M$

Hence, M is skew symmetric

S2: $Let, N = A^{26}C^{13} - C^{13}A^{26}$

Then $N^T = \left(A^{26}C^{13}\right)^T - \left(C^{13}A^{26}\right)^T$

$= -(C)^{13}(A)^{26} + A^{26}C^{13} = N$

Hence, N is symmetric.

$\therefore$ Only S2 is true.

53. (4)

$$T_{r+1} = {}^{22}C_r \cdot \left(x^m\right)^{22-r} \cdot \left(\frac{1}{x^2}\right)^r$$

$$T_{r+1} = {}^{22}C_r \cdot x^{22m-m-2r}$$

For coefficient of x^1

$\because \quad 22m - mr - 2r = 1$ and ${}^{22}C_r = 1540$

$\because \quad {}^{22}C_3 = 1540 \Rightarrow r = 3$ or 19

Now, for $r = 3; 22m - 3m - 6 = 1$

$\Rightarrow \quad 19m = 7 \Rightarrow m = \frac{7}{19}$

for $r = 19; 22m - 19m = 39 \Rightarrow m = 13$

54. (2)

$$-\frac{1}{5} - \frac{3}{5}$$

$$z = \frac{(1+i)^2}{a-i}$$

On rationalising the denominator, we get

$$z = \frac{2ai - 2}{a^2 + 1}$$

$$\Rightarrow |z| = \frac{2}{\sqrt{a^2+1}} = \sqrt{\frac{2}{5}}$$

$$\Rightarrow 20 = 2\left(a^2 + 1\right)$$

$$\Rightarrow a^2 + 1 = 10$$

$$\Rightarrow a = 3 \quad (\because a > 0)$$

$$\therefore z = \frac{2(3)i - 2}{3^2 + 1}$$

$$z = \frac{-1 + 3i}{5}$$

$$\Rightarrow \overline{z} = \frac{-1 - 3i}{5}$$

55. (4)

$L_1 : 3x - 4y + 12 = 0$

$L_2 : 8x + 6y + 11 = 0$

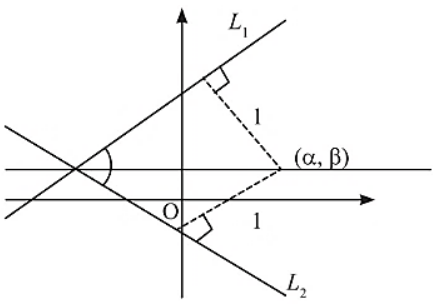
(α, β) lies on that angle which contain origin

$\therefore$ Equation of angle bisector of that angle which contain origin is

$$\frac{3x - 4y + 12}{5} = \frac{8x + 6y + 11}{10} \Rightarrow 2x + 14y - 13 = 0$$

$$\Rightarrow \quad 2\alpha + 14\beta - 13 = 0 \quad \ldots(i)$$

Again $\frac{3\alpha - 4\beta + 12}{5} = 1 \ldots(ii)$



$$\Rightarrow \quad 3\alpha - 4\beta + 7 = 0$$

Solving (i) and (ii)

$$\alpha = -\frac{23}{25} \text{ and } \beta = \frac{53}{50}$$

$$\therefore \quad \alpha + \beta = \frac{7}{50}$$

$$\Rightarrow \ 100(\alpha + \beta) = 14$$

56. (1)

On rearranging, we get $\frac{x+1}{1} = \frac{y}{\frac{1}{2}} = \frac{z}{\frac{-1}{12}}$

and $\frac{x}{1} = \frac{y+2}{1} = \frac{z-1}{\frac{1}{6}}$

Shortest distance between the lines (S.D.)

$$= \frac{(\vec{b} - \vec{a}) \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}$$

Here $\vec{a} = -\hat{i} \quad \vec{b} = -2\hat{j} + \hat{k}$

$$\vec{p} = \hat{i} + \frac{\hat{j}}{2} + \left(\frac{-\hat{k}}{12}\right)$$

$$\vec{q} = \hat{i} + \hat{j} + \frac{\hat{k}}{6}$$

$$S.D. = (\hat{i} - 2\hat{j} + \hat{k}) \cdot \frac{(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}$$

$$\therefore \vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \frac{1}{2} & \frac{-1}{12} \\ 1 & 1 & \frac{1}{6} \end{vmatrix} = \frac{1}{6}\hat{i} - \frac{1}{4}\hat{j} + \frac{1}{2}\hat{k}$$

$$= 2\hat{i} - 3\hat{j} + 6\hat{k}$$

$$\text{S.D.} = \left|\frac{(\hat{i} - 2\hat{j} + \hat{k}) \cdot (2\hat{i} - 3\hat{j} + 6\hat{k})}{\sqrt{2^2 + 3^2 + 6^2}}\right| = \left|\frac{14}{7}\right| = 2$$

57. (4)

Class	Frequency	x_i	$x_i f_i$
0-6	a	3	$3a$
6-12	b	9	$9b$
12-18	12	15	180
18-24	9	21	189
24-30	5	27	135
	$N = a + b + 26$		

$$\text{Mean } x = \frac{\sum x_i f_i}{N}$$

$$\Rightarrow \ \frac{309}{22} = \frac{3a + 9b + 504}{a + b + 26}$$

$$\Rightarrow \ 81a + 37b = 1018 \ \ldots(i)$$

$$\text{For Median} = L + \frac{\frac{N}{2} - c}{f} \times h$$

$$14 = 12 + \frac{\frac{a+b}{2} + 13 - (a+b)}{12} \times 6$$

$$\Rightarrow \ a + b = 18 \qquad \ldots(ii)$$

By solving equation (i) and equation (ii), we get $a = 8$ and $b = 10 \Rightarrow (a-b)^2 = 4$

58. (1)

$S_4 = 16$

Let first term = a and common difference = d

$$2\{2a + 3d\} = 16 \quad \left[\because s_n = \frac{n}{2}[2a + (n-1)d]\right]$$

$S_6 = -48$

$3(2a + 5d) = -48$

So, $2a + 3d = 8 \qquad \ldots(i)$

$2a + 5d = -16 \qquad \ldots(ii)$

Solve (i) and (ii)

$\Rightarrow \quad d = -12$

So, $S_{10} = 5\{44 - 9 \times 12\} = -320$

59. (3)

$y^2 = 8x + 4y + 4$

$(y-2)^2 = 8(x + 1)$

$Y^2 = 4aX$

where, $a = 2$, $X = x + 1$, $Y = y-2$

focus $\equiv (2, 0)$ and $x + 1 = 2$, $y - 2 = 0$

focus $(1, 2)$

$y-2 = m(x-1)$

Put (3,0) in the above line

$\Rightarrow \quad m = -1, \theta = 3\pi/4$

Length of focal chord $= 16 = 4a \operatorname{cosec}^2 \theta$

60. (2)

$$\int_{\cos x}^{1} t^2 f(t)dt = \sin^3 x + \cos x - 1$$

Differentiating both sides we get,

$$-\cos^2 x f(\cos x) \cdot (-\sin x) = 3\sin^2 x \cdot \cos x - \sin x$$

$$\Rightarrow \quad f(\cos x) = 3\tan x - \sec^2 x$$

$$\Rightarrow \quad f'(\cos x)(-\sin x) = 3\sec^2 x - 2\sec^2 x \tan x$$

$$\Rightarrow \quad f'(\cos x)\cos x = \frac{2}{\cos^2 x} - \frac{3}{\sin x \cdot \cos x}$$

$$\text{When } \cos x = \frac{1}{\sqrt{3}}; \sin x = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\therefore \quad f'\left(\frac{1}{\sqrt{3}}\right)\frac{1}{\sqrt{3}} = 6 - \frac{9}{\sqrt{2}}.$$

61. **(1)**

5	x	x	x	0
7	x	x	x	0
7	x	x	x	5
9	x	x	x	0
9	x	x	x	5

So, required numbers $=5\times {}^4C_3 \times 3! = 120$

62. **(4)**

We have, $4x^2+4y^2-12x+8y+k=0$

$\because$ $\left(1,-\frac{1}{3}\right)$ lies on or inside the C then

$$4+\frac{4}{9}-12-\frac{8}{3}+k\le 0\Rightarrow k\le\frac{92}{9}$$

Now, circle lies in 4th quadrant with centre

$\equiv\left(\frac{3}{2},-1\right)$

$\therefore \quad r<1\Rightarrow\sqrt{\frac{9}{4}+1-\frac{k}{4}}<1$

$\Rightarrow \quad \frac{13}{4}-\frac{k}{4}<1\Rightarrow\frac{k}{4}<\frac{9}{4}\Rightarrow k>9$

$\therefore \quad k\in\left(9,\frac{92}{9}\right]$

63. **(1)**

Let $f(x)=x^2+2(a+4)x-5a+64$

For $f(x)>0$

We have D < 0

$\Rightarrow \quad 4\left[(a+4)^2-(64-5a)\right]<0$



$\Rightarrow \quad a^2+13a-48<0$

$\Rightarrow \quad (a+16)(a-3)<0$

$\Rightarrow \quad -16<a<3$

$\because \quad a\in[-5,30]$

$\Rightarrow \quad -5\le a\le 3$

$\Rightarrow \quad a=-5,-4,\ldots..,1,2$ (integers)

$\therefore \quad n(E)=8$ and $n(S)=36$

Probability $P(E)=\frac{8}{36}=\frac{2}{9}$

64. **(1)**

$kx^2+k^2y^2=1$

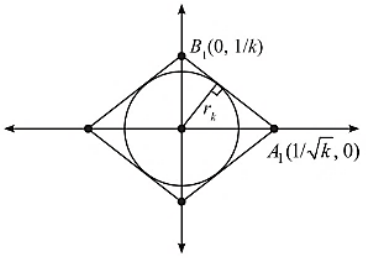
$\frac{x^2}{1/k}+\frac{y^2}{1/k^2}=1$

Now

Equation of

$A_1B_2;\frac{x}{1/\sqrt{k}}+\frac{y}{1/k}=1\Rightarrow\sqrt{k}x+ky=1$

$r_i=\perp$ distance of $(0,0)$ from line A_1B_1



$$r_k=\left|\frac{(0+0-1)}{\sqrt{k+k^2}}\right|=\frac{1}{\sqrt{k+k^2}}$$

$$\frac{1}{r_k^2}=k+k^2\Rightarrow\sum_{k=1}^{20}\frac{1}{r_k^2}=\sum_{k=1}^{20}\left(k+k^2\right)$$

$$=\frac{\sum_{k=1}^{20}k+\sum_{k=1}^{20}k^2}{2}$$

$$=\frac{20\times 21}{2}+\frac{20.21.41}{6}$$

$=210 + 10 \times 7 \times 41$

$=210 + 2870$

$= 3080$

65. **(4)**

$\vec{m}_1$ and $\vec{m}_2$ are normal vector to the plane $\hat{i}+\hat{j},\hat{i}+\hat{k}$ and $\hat{i}-\hat{j};\hat{j}-\hat{k}$

$$\vec{m}_1=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&0\\1&0&1\end{vmatrix}=\hat{i}-\hat{j}-\hat{k}$$

$$\vec{m}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&0\\0&1&-1\end{vmatrix}=\hat{i}+\hat{j}+\hat{k}$$

$\vec{a}=\mu\left|\vec{m}_2\times\vec{m}_1\right|$

$$\vec{a}=\mu\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&-1\\1&1&1\end{vmatrix}=\mu(-2\hat{j}+2\hat{k})$$

So, $\vec{a}\cdot\vec{b}=\mu(0+4+2)=6$

$[\because \vec{a}\cdot\vec{b}=6]$

$\Rightarrow \quad \mu=1$

So, $\vec{a}=-2\hat{j}+2\hat{k}$

Now, $\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

$\Rightarrow \quad \cos\theta=\frac{6}{2\sqrt{2}\times 3}=\frac{1}{\sqrt{2}}$

$\Rightarrow \quad \theta=\frac{\pi}{4}$

Now, $\vec{a}\cdot\vec{b}|^2+|\vec{a}\times\vec{b}|^2=|\vec{a}|^2|\vec{b}|^2$

$\Rightarrow \quad 36+|\vec{a}\times\vec{b}|^2=8\times 9=72$

$\Rightarrow \quad |\vec{a}\times\vec{b}|^2=36$

$\Rightarrow |\vec{a}\times\vec{b}|=6$

66. (3)

Let $P=\left(t_1^2,2t_1\right)$ and $Q=\left(t_2^2,2t_2\right)$

Equation of PQ is $(t_1+t_2)\,y=2x+2t_1t_2$

$\Rightarrow$ lines $-2x+(t_1+t_2)\,y=2t_1t_2$ and $lx-2y=1$ are same

$\Rightarrow \quad \frac{-2}{l}=\frac{t_1+t_2}{-2}=\frac{2t_1t_2}{1}$

$\Rightarrow \quad t_1+t_2=\frac{4}{l}\ \&\ t_1t_2=\frac{-1}{l}$

Now, $R=\left(\frac{1}{t_1^2},\frac{-2}{t_1}\right)$ and $T=\left(\frac{1}{t_2^2},\frac{-2}{t_2}\right)$

The slope of RT is $\frac{\frac{-2}{t_2}+\frac{2}{t_1}}{\frac{1}{t_2^2}-\frac{1}{t_1^2}}=\frac{-\frac{2(t_1-t_2)}{t_1t_2}}{\frac{(t_1-t_2)(t_1+t_2)}{t_1^2t_2^2}}$

$=\frac{-2t_1t_2}{t_1+t_2}=\frac{\frac{2}{l}}{\frac{4}{l}}=\frac{1}{2}=0.5$

67. (3)

$2(y+2)\ln(y+2)dx+(x+4-2\ln(y+2))dy=0$

$2\ln(y+2)+(x+4-2\ln(y+2))\frac{1}{y+2}\cdot\frac{dy}{dx}=0$

Let, $\log_e(y+2)=t$, then $\frac{1}{y+2}\cdot\frac{dy}{dx}=\frac{dt}{dx}$

Hence, $2t+(x+4-2t)\cdot\frac{dt}{dx}=0$

$(x+4-2t)\frac{dt}{dx}=-2t$

$\Rightarrow \quad \frac{dx}{dt}=\frac{2t-4-x}{2t} \Rightarrow \frac{dx}{dt}+\frac{x}{2t}=\frac{2t-4}{2t}$

$I\cdot F=e^{\int\frac{1}{2t}dt}=t^{\frac{1}{2}}$

$x\cdot t^{1/2}=\int\frac{2t-4}{2t}\cdot t^{1/2}\cdot dt$

$x\cdot t^{1/2}=\int\left(t^{1/2}-\frac{2}{t^{1/2}}\right)\cdot dt$

$=\frac{t^{\frac{3}{2}}}{\frac{3}{2}}-2\cdot\frac{t^{\frac{1}{2}}}{\frac{1}{2}}+C$

$x\cdot t^{\frac{1}{2}}=\frac{2t^{\frac{3}{2}}}{3}-4t^{\frac{1}{2}}+C$

$x=\frac{2}{3}\cdot t-4+C\cdot t^{\frac{-1}{2}}$

$x=\frac{2}{3}\ln(y+2)-4+\ \text{C.}\ (\ln(y+2))^{\frac{-1}{2}}$

Put $y=e^4-2, x=1$

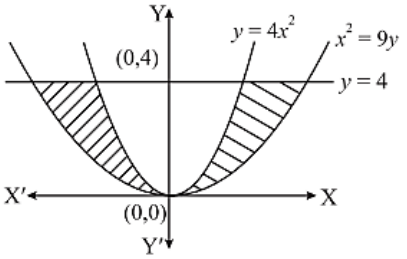
$\Rightarrow 1=\frac{2}{3}\times4-4+C\times\frac{1}{2}$

$\Rightarrow \frac{C}{2}=5-\frac{8}{3}=\frac{7}{3} \Rightarrow C=\frac{14}{3}$

$x=\frac{2}{3}\times9-4+\frac{14}{3}\times\frac{1}{3}$ [put $x=e^9-2$]

$=2+\frac{14}{9}=\frac{32}{9}$

68. (4)



Required area $=2\cdot\int_0^4\left(3\sqrt{y}-\frac{\sqrt{y}}{2}\right)dy$

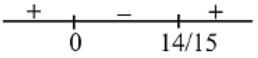
$=2\cdot\int_0^4\frac{5}{2}\sqrt{y}dy=5\cdot\left[\frac{y^{3/2}}{\frac{3}{2}}\right]_0^4=\frac{80}{3}$

69. (1)

$f(x)=(3x-7)x^{2/3}$

$\Rightarrow \quad f(x)=3x^{5/3}-7x^{2/3}$

$\Rightarrow \quad f'(x)=5x^{2/3}-\frac{14}{3x^{1/3}}$



For increasing function $=\frac{15x-14}{3x^{1/3}}>0$

$\therefore \quad f'(x)>0\forall x\in(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$

70. (3)

$7^{2022}+3^{2022}$

$=(49)^{1011}+(9)^{1011}$

$(50-1)^{1011}+(10-1)^{1011}$

$=k_1-1+5k_2-1$

$=5m-2$

Remainder $=5-2=3$

71. (2)

After rationalising

$$=\frac{1}{\alpha}\int_0^{\alpha}\left[x\sqrt{x+\alpha}+x^{3/2}\right]dx$$

$$=\frac{1}{\alpha}\left[x(x+\alpha)^{3/2}\cdot\frac{2}{3}-\int(x+\alpha)^{3/2}\cdot\frac{2}{3}dx+x^{5/2}\cdot\frac{2}{5}\right]_0^{\alpha}$$

$$=\frac{1}{\alpha}\left[\frac{2}{3}x(x+\alpha)^{3/2}-\frac{2}{3}\times\frac{2}{5}(x+\alpha)^{5/2}+\frac{2}{5}x^{5/2}\right]_0^{\alpha}$$

$$=\frac{1}{\alpha}\left[\frac{2}{3}\alpha(2\alpha)^{3/2}-\frac{4}{15}(2\alpha)^{5/2}+\frac{2}{5}\alpha^{5/2}+\frac{4}{15}\alpha^{5/2}\right]$$

$$=\frac{1}{\alpha}\cdot\alpha^{5/2}\left[\frac{4\sqrt{2}}{3}-\frac{16\sqrt{2}}{15}+\frac{10}{15}\right]$$

$$=\alpha^{3/2}\left[\frac{4\sqrt{2}+10}{15}\right]=\frac{16+20\sqrt{2}}{15}$$

$$\Rightarrow\ \alpha^{3/2}\left[\frac{4\sqrt{2}+10}{15}\right]=2\sqrt{2}\left(\frac{4\sqrt{2}+10}{15}\right)$$

$$\Rightarrow\ \alpha=2$$

72. (3)

Let $\tan\theta=\frac{\pi}{\sqrt{2}}\Rightarrow\frac{\pi}{4}\le\theta\le\frac{\pi}{2}\Rightarrow\pi=\sqrt{2}\tan\theta$

$$\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+2\tan^2\theta}}+\frac{1}{4}\sin^{-1}$$

$$\left(\frac{2\sqrt{2}\sqrt{2}\tan\theta}{2+2\tan^2\theta}\right)+\tan^{-1}\left(\frac{\sqrt{2}}{\sqrt{2}\tan\theta}\right)$$

$$=\frac{3}{2}\cos^{-1}(\cos\theta)+\frac{1}{4}\sin^{-1}(\sin 2\theta)+\tan^{-1}(\cot\theta)$$

$$\frac{3}{2}\theta+\frac{1}{4}(\pi-2\theta)+\frac{\pi}{2}-\theta$$

$$=\frac{\pi}{4}+\frac{\pi}{2}=\frac{3\pi}{4}$$

73. (64)

$$I(x)=\int\sqrt{\frac{x+7}{x}}dx$$

Put $x=\mu^2\quad dx=2\mu d\mu$

$$\int 2\sqrt{\mu^2+7}d\mu=2\int\sqrt{\mu^2+\left(\sqrt{7}^2\right)}d\mu$$

$$I(\mu^2)=2\left[\frac{\mu}{2}\sqrt{\mu^2+7}+\frac{7}{2}\ln\left|\mu+\sqrt{\mu^2+7}\right|\right]+C$$

Put $\mu=\sqrt{x}$

$I(x)=\sqrt{x}\sqrt{x+7}+7\ln|\sqrt{x}+\sqrt{x+7}|+C$

$I(9)=12+7\ln 7=12+7(\ln(3+4))+C$

$\Rightarrow\ C=0\ I(x)=\sqrt{x}\sqrt{x+7}+7\ln(\sqrt{x}+\sqrt{x+7})$

$I(1)=1\sqrt{8}+7\ln(1+\sqrt{8})$

$I(1)=\sqrt{8}+7\ln(1+2\sqrt{2})$

$=\alpha+7\log_c(1+2\sqrt{2})$

So, $\alpha=\sqrt{8}$

$\Rightarrow\ \alpha^4=\left(8^{1/2}\right)^4$

$\Rightarrow\ \alpha^4=8^2=64$

74. (6)

$A=\{1, 2, 3, 4\}$

$R=\{(a, b), (c, d)\}$

$2a + 3b = 4c + 5d = \beta$ let

$2a=\{2, 4, 6, 8\}$ $4c=\{4, 8, 12, 16\}$

$3b = \{3, 6, 9, 12\}5d(5, 10, 15, 20\}$

$$2a+3b=\begin{Bmatrix}5,8,11,14\\7,10,13,16\\9,12,15,18\\11,14,17,20\end{Bmatrix}4c+5d\begin{Bmatrix}9,14,19,24\\13,18\ldots\\17,22\ldots\\21,26\ldots\end{Bmatrix}$$

Possible value of $\beta = 9, 13, 14, 14, 17, 18$

Then number of elements in R is 6.

75. (3)

Given $16\sin 20^\circ\sin 40^\circ\sin 80^\circ$

$=16\sin 40^\circ\sin 20^\circ\sin 80^\circ$

$4(4\sin(60-20)\sin(20)\sin(60+20))$

$=4\sin\left(3\times 20^\circ\right)$

$\therefore\sin 30=4\sin(60-\theta)\sin\theta\sin(60+\theta)]$

$=4\times\sin 60^\circ$

$=4\times\frac{\sqrt{3}}{2}=2\sqrt{3}$



PW Web/App - https://smart.link/7wwosivoicgd4

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